# Xiaoyang Xu

xiaoyangxu@cs.ucsb.edu | xyxu2033.github.io | Google Scholar

## Education

**University of California, Santa Barbara** Sep. 2024–present
Ph.D. student in Computer Science; advisor: Daniel Lokshtanov.

**University of Science and Technology of China** Sep. 2020–Jun. 2024
B.Eng. (Honors) in Computer Science, School of the Gifted Young.

## Selected Honors and Awards

**Outstanding Poster Award**, ICBS College Poster Session; *Efficiency of Pairing Heaps*. 2026
**GUO Moruo Scholarship**, the highest honor at USTC. 2024
**Gold Award**, ICPC China National Invitational Programming Contest (Kunming). 2024
**China National Scholarship**, Ministry of Education. 2023
**Selected Participant**, 10th Heidelberg Laureate Forum. 2023

## Service

**Sub-reviewer:** SODA 2027 (2×), SOSA 2027, IWOCA 2026, ITCS 2026, ESA 2025.
**Co-coach**, UC Santa Barbara Competitive Programming Teams. Nov. 2024–present
**Teaching assistant:**

- UCSB CMPSC 130A: Data Structures and Algorithms I (Spring 2026, Fall 2024).
- UCSB CMPSC 132: Algorithmic Engineering (Fall 2025).
- UCSB CMPSC 130B: Data Structures and Algorithms II (Spring 2025, Winter 2025).
- USTC 011186: Algorithms for Big Data (Spring 2023).

## Academic Visits

**New York University Shanghai**, with Jie Xue. Sep. 2026
**University of Bergen**, with Fedor V. Fomin. Feb.–Mar. 2026
**Toyota Technological Institute at Chicago**, with Santhoshini Velusamy. Jun.–Sep. 2025
**Harvard University**, with Heng Yang. Jul. 2023–Jan. 2024

## Publications

[1] **Efficiency of the Pairing Heap.**
**Xiaoyang Xu**, Robert E. Tarjan. *SODA 2027*, to appear.

[2] **Pure Pairing Heaps.**
Robert E. Tarjan, **Xiaoyang Xu**. In submission, 2026.

[3] **Learning-Augmented Streaming Algorithms for Approximating Boolean Max-CSPs.**
Yinhao Dong, Pan Peng, Ali Vakilian, Santhoshini Velusamy, **Xiaoyang Xu**. *NeurIPS 2026*, to appear.

[4] **Parameterized Approximation of Rectangle Stabbing.**
Huairui Chu, Ajaykrishnan E S, Daniel Lokshtanov, Anikait Mundhra, Thomas Schibler, **Xiaoyang Xu**, Jie Xue. *ESA 2026*.

[5] **Fast and Certifiable Trajectory Optimization.**
Shucheng Kang, **Xiaoyang Xu**, Jay Sarva, Ling Liang, Heng Yang. *WAFR 2024*.

[6] **A Novel Skip Orthogonal List for Dynamic Optimal Transport Problem.**
**Xiaoyang Xu**, Hu Ding. *AAAI 2024*.

## Talks and Posters

**Efficiency of Pairing Heaps**

- Theory Lunch, New York University Shanghai, September 16, 2026.
- ICBS 2026 College Poster Session. [Poster]

**Learning-Augmented Streaming Algorithms for Approximating Max-DICUT**

- Algorithms Seminar, University of Bergen, March 12, 2026. [Slides]

**Skip Orthogonal List for Dynamic Optimal Transport**

- AAAI Conference on Artificial Intelligence, 2024. [Poster]